LAB 2 — HOMEWORK

Data Types and Console I/O

C# Programming — Take-home Assignment

Variables, Parsing, and Formatted Console Output

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Course: C# Programming

1. Overview

This take-home assignment accompanies Lab 2 — Data Types and Console I/O. It contains three exercises that reinforce what was covered in class: declaring variables with primitive data types, reading input from the keyboard, converting strings to numeric types, and producing well-formatted console output.

The exercises are arranged from fully guided to fully independent:

- **Exercise 1** — fully guided, with step-by-step code and explanations.
- **Exercise 2** — partial hints only; you write the complete program.
- **Exercise 3** — no hints. Apply everything you have learned.

1.1 Required Knowledge

Before starting, make sure you are comfortable with the following topics from Lab 2:

- Declaring and initializing variables (int, double, decimal, string, bool, char).
- Using const for values that must not change.
- Reading input with Console.ReadLine().
- Parsing strings with int.Parse, double.Parse, decimal.Parse, and the safer TryParse variants.
- Printing output with Console.Write, Console.WriteLine, string interpolation ($"..."), and composite formatting ("{0}", "{1}").
- Numeric format specifiers: N2, F2, C, P, D, and field width/alignment such as {x,-10} or {y,10:N2}.

1.2 Submission Requirements

You must submit BOTH of the following for this assignment:

1. A single Word document (.docx) containing your answers, screenshots of the program running, and an explanation of your approach for each exercise.
2. The full source code of all three programs, with each exercise as a separate Visual Studio project. Zip the three project folders together into one archive before submitting.

File naming and packaging

- Name the Word document: StudentID_FullName_Lab02_Homework.docx
- Name the source archive: StudentID_FullName_Lab02_Homework.zip
- The .zip must contain three folders — one per exercise — each with its own .sln/.csproj. Example structure:

```
StudentID_FullName_Lab02_Homework.zip
├── Ex01_GreetingCard/
│   ├── Ex01_GreetingCard.sln
│   └── Ex01_GreetingCard/ ...
```

```
├── Ex02_RectangleStats/
│   ├── Ex02_RectangleStats.sln
│   └── Ex02_RectangleStats/ ...
└── Ex03_ElectricityBill/
    ├── Ex03_ElectricityBill.sln
    └── Ex03_ElectricityBill/ ...
```

Project rules

- Each exercise MUST be created as its own separate Console App (C#) project in Visual Studio 2022, targeting .NET 8.0. Do not put all three exercises in one single project.
- For each project, name the project exactly as shown in the structure above.
- If your program prints Vietnamese characters, set Console.OutputEncoding = System.Text.Encoding.UTF8; at the start of Main().
- Clean, consistent indentation and meaningful variable names are expected.

What to include in the Word document

- A cover page with your full name, student ID, class, and the date.
- For each exercise: the problem statement (you may copy it from this document), your source code (pasted as text — not only a screenshot), at least one screenshot showing the program running with real input, and a short paragraph (3–5 sentences) explaining what you did and any difficulties you encountered.

Exercise 1 — Personal Greeting Card (Guided)

Project name: Ex01_GreetingCard

Problem statement

Write a C# console program that asks the user for four pieces of personal information: full name, year of birth, home city, and current GPA. The program then computes the user's age (based on the current year) and prints a nicely formatted greeting card.

Sample run

Example interaction:

```
Enter your full name: Nguyen Van A
Enter your year of birth: 2004
Enter your home city: Ho Chi Minh City
Enter your GPA (0.0 - 4.0): 3.25

========================================
             GREETING CARD
========================================
Name      : Nguyen Van A
Age       : 22
City      : Ho Chi Minh City
GPA       : 3.25
========================================
Hello Nguyen Van A, welcome to C#!
```

Step-by-step guide

Follow the steps below. The full code is provided at the end of the guide.

Step 1 — Create the project

- Open Visual Studio 2022. Choose File → New → Project → Console App (C#).
- Name the project Ex01_GreetingCard and pick .NET 8.0 as the framework.
- Open Program.cs and remove the default Console.WriteLine("Hello, World!"); line.

Step 2 — Set up UTF-8 output

To make sure Vietnamese characters display correctly, add this line as the first statement inside Main():

```
Console.OutputEncoding = System.Text.Encoding.UTF8;
```

Step 3 — Read the four inputs

Use Console.Write to display the prompt on the same line as the input, and Console.ReadLine() to read the user's text. Because name and city are strings, they do not need to be converted. Year of birth must be converted to int, and GPA to double.

```
Console.Write("Enter your full name: ");
string name = Console.ReadLine();

Console.Write("Enter your year of birth: ");
int year = int.Parse(Console.ReadLine());

Console.Write("Enter your home city: ");
string city = Console.ReadLine();

Console.Write("Enter your GPA (0.0 - 4.0): ");
double gpa = double.Parse(Console.ReadLine());
```

Step 4 — Compute the age

The DateTime.Now property gives the current date. Its Year property returns the current year as an int. Subtract the user's year of birth to get the age.

```
int age = DateTime.Now.Year - year;
```

Step 5 — Print the formatted card

Use new string('=', 40) to repeat the '=' character 40 times. Use string interpolation with field width ({x,-10}) to left-align the labels so the values line up into a neat column.

```
Console.WriteLine();
Console.WriteLine(new string('=', 40));
Console.WriteLine("             GREETING CARD");
Console.WriteLine(new string('=', 40));
Console.WriteLine($"{\"Name\",-10}: {name}");
Console.WriteLine($"{\"Age\",-10}: {age}");
Console.WriteLine($"{\"City\",-10}: {city}");
Console.WriteLine($"{\"GPA\",-10}: {gpa:F2}");
Console.WriteLine(new string('=', 40));
Console.WriteLine($"Hello {name}, welcome to C#!");
```

Step 6 — Run and test

- Press F5 to run the program.
- Try several inputs: a normal case, a long name, a year that makes age very small (for example 2024 → age 2).
- Take a screenshot of the output to include in your Word document.

Complete reference code

```
using System;

namespace Ex01_GreetingCard
{
```

```
    class Program
    {
        static void Main()
        {
            Console.OutputEncoding = System.Text.Encoding.UTF8;

            Console.Write("Enter your full name: ");
            string name = Console.ReadLine();

            Console.Write("Enter your year of birth: ");
            int year = int.Parse(Console.ReadLine());

            Console.Write("Enter your home city: ");
            string city = Console.ReadLine();

            Console.Write("Enter your GPA (0.0 - 4.0): ");
            double gpa = double.Parse(Console.ReadLine());

            int age = DateTime.Now.Year - year;

            Console.WriteLine();
            Console.WriteLine(new string('=', 40));
            Console.WriteLine("             GREETING CARD");
            Console.WriteLine(new string('=', 40));
            Console.WriteLine($"{\"Name\",-10}: {name}");
            Console.WriteLine($"{\"Age\",-10}: {age}");
            Console.WriteLine($"{\"City\",-10}: {city}");
            Console.WriteLine($"{\"GPA\",-10}: {gpa:F2}");
            Console.WriteLine(new string('=', 40));
            Console.WriteLine($"Hello {name}, welcome to C#!");
        }
    }
}
```

Your task

Type (do NOT copy-paste) the code above into your own project, run it, and paste both the source code and a screenshot of the run into your Word document. After the screenshot, add a short paragraph (3–5 sentences) explaining what each of steps 3, 4, and 5 is doing in your own words.

Exercise 2 — Rectangle Statistics (Hints)

Project name: Ex02_RectangleStats

Problem statement

Write a C# console program that reads the width and height of a rectangle (both in meters) from the keyboard, then computes and prints: the perimeter, the area, the length of a diagonal, and the area converted to square centimeters. All measurements must use the double type. Results must be printed with exactly two decimal places, using string interpolation.

Sample run

Example interaction:

```
=== Rectangle Statistics ===
Enter the width  (m): 3.5
Enter the height (m): 2.0

Width     :      3.50 m
Height    :      2.00 m
Perimeter :     11.00 m
Area      :      7.00 m^2
Diagonal  :      4.03 m
Area      : 70,000.00 cm^2
```

Requirements

- Use double for all measurements (not int or decimal).
- Use double.Parse to convert each input from string to double.
- Use the Math.Sqrt method to compute the diagonal.
- Display every numeric value with exactly 2 decimal places. For the square-centimeter value, also include a thousands separator (see hints below).
- Print a title line and a blank line between the inputs and the results, exactly as shown in the sample output.

Hints

Hint 1 — Formulas

- perimeter = 2 × (width + height)
- area = width × height
- diagonal = $\sqrt{(width^2 + height^2)}$ — use Math.Sqrt(width * width + height * height)
- 1 m^2 = 10,000 cm^2, so areaCm2 = area × 10000

Hint 2 — Reading a double

Remember that Console.ReadLine() returns a string. You must convert it to double:

```
Console.Write("Enter the width  (m): ");
double width = double.Parse(Console.ReadLine());
```

Hint 3 — Format specifiers

Two decimal places without a thousands separator is F2. Two decimal places WITH a thousands separator is N2. Field width is written as {value,10:F2} (right-aligned in 10 character columns). Example:

```
Console.WriteLine($"{\"Area\",-10}: {area,9:F2} m^2");
Console.WriteLine($"{\"Area\",-10}: {areaCm2,9:N2} cm^2");
```

Hint 4 — Program skeleton

Fill in the blanks and complete the program:

```
using System;

namespace Ex02_RectangleStats
{
    class Program
    {
        static void Main()
        {
            Console.OutputEncoding = System.Text.Encoding.UTF8;
            Console.WriteLine("=== Rectangle Statistics ===");

            // TODO 1: read width (double) from the keyboard
            // TODO 2: read height (double) from the keyboard

            // TODO 3: compute perimeter, area, diagonal, areaCm2

            Console.WriteLine();
            // TODO 4: print each value using string interpolation
            //         with F2 for metres and N2 for square centimetres

            // Example line you can reuse:
            // Console.WriteLine($"{\"Area\",-10}: {area,9:F2} m^2");
        }
    }
}
```

Your task

- Create the project Ex02_RectangleStats and implement the program. You are expected to write the complete working code yourself — the skeleton is only a starting point.

- Test it with at least three different input pairs, including one with decimal values such as 3.5 and 2.0.
- In your Word document, paste the full source code, add one screenshot of the running program, and write 3–5 sentences explaining how you computed the diagonal and why F2 was used for metres and N2 was used for square centimetres.

Exercise 3 — Electricity Bill Calculator (Independent)

Project name: Ex03_ElectricityBill

Problem statement

Write a C# console program that calculates a simplified residential electricity bill. The program must:

1. Ask the customer to enter their full name and the number of kWh they consumed this month. The name is a string; consumption is an integer (kWh).
2. Compute the electricity charge using the tiered price table below.
3. Compute a 10% VAT on the electricity charge.
4. Print a formatted bill that shows the customer name, the current date, the consumption, the electricity charge, the VAT, and the grand total.

Tiered price table

Tier	Consumption range (kWh)	Unit price (VND/kWh)
Tier 1	First 50 kWh	1,800
Tier 2	Next 50 kWh (51-100)	2,000
Tier 3	Next 100 kWh (101-200)	2,500
Tier 4	Above 200 kWh	3,000

Example: if the customer consumed 230 kWh, the electricity charge is:

- First 50 kWh × 1,800 = 90,000 VND
- Next 50 kWh × 2,000 = 100,000 VND
- Next 100 kWh × 2,500 = 250,000 VND
- Last 30 kWh × 3,000 = 90,000 VND
- Total before VAT = 530,000 VND; VAT = 53,000 VND; Grand total = 583,000 VND.

Required output format

Your program's output must look similar to this:

```
================ ELECTRICITY BILL ================
Customer    : Nguyen Van A
Bill date   : 22/04/2026
Consumption:       230 kWh
--------------------------------------------------
Charge      :   530,000.00 VND
VAT (10%)   :    53,000.00 VND
```

```
--------------------------------------------------
TOTAL      :   583,000.00 VND
==================================================
```

Requirements

- Each exercise must be its own separate Visual Studio project. Do NOT reuse Ex01 or Ex02.
- All monetary values must use the decimal type (not double), because money must not lose precision.
- The consumption value must be an int.
- Display the bill date using DateTime.Now with the dd/MM/yyyy format.
- Use the N2 format specifier for every monetary value.
- Use string.Format, string interpolation, or composite formatting with field width and alignment to make the columns line up neatly, just like the sample output.
- You may declare the four tier prices as const decimal values at the top of Main() (recommended, but not mandatory).
- Do NOT use any C# features that have not been covered in Lab 2 (in particular: do not use if/else, loops, arrays, methods other than Main, or classes other than Program — those will be covered later). All of the tier logic can be expressed with simple arithmetic using the Math.Max / Math.Min functions, or by breaking the consumption into parts step by step with straightforward subtraction.

What to submit for this exercise

- The complete source code pasted into the Word document.
- At least three screenshots, each showing a run with a different consumption value — one below 50 kWh, one between 100 and 200 kWh, and one above 200 kWh.
- A short paragraph (5–8 sentences) explaining: why you chose decimal instead of double for the money values; how your code computes the charge for each tier; and one difficulty you ran into and how you solved it.